ClipSAT : Egyptian Scholastic Test: EST II Mathematics Level 2

Unit 4.01: Matrices, Determinants & 2D/3D Vectors

Name: _____

Q1. A vector mathbf{v} is represented by the directed line segment from point P1, 2 to point Q4, 6 . What are the components of vector mathbf{v} ?

[ ] A: [3, 4]
[ ] B: [5, 8]
[ ] C: [-3, -4]
[ ] D: [4, 3]

Q2. A translation maps point A1, 2 to A'-4, 2 and point B3, 5 to B'-2, 5 . This transformation is represented by begin{pmatrix} x' y' end{pmatrix} = begin{pmatrix} x y end{pmatrix} + begin{pmatrix} a b end{pmatrix} . Find the values of a and b .

[ ] A: a = -5, b = 0
[ ] B: a = 5, b = 0
[ ] C: a = 0, b = -5
[ ] D: a = 0, b = 5

Q3. A linear transformation in the 2D plane is represented by the matrix M = begin{pmatrix} a & b c & d end{pmatrix} . This transformation maps the point P1, 2 to P'-3, 7 and the point Q3, 1 to Q'5, 1 . Determine the values of a, b, c, and d , and then calculate the sum a+b+c+d .

[ ] A: 9
[ ] B: 11
[ ] C: 7
[ ] D: 5

Q4. A matrix M = begin{pmatrix} k & 2 3 & 1 end{pmatrix} transforms a point P1, 2 to P'5, 5 . Find the value of k .

[ ] A: 1
[ ] B: 2
[ ] C: 3
[ ] D: 4

Q5. Consider the matrix A = begin{pmatrix} 2 & 0 0 & 3 end{pmatrix} . When the matrix A is applied to the vector mathbf{v} = begin{pmatrix} x y end{pmatrix} , the resulting vector is Amathbf{v} = begin{pmatrix} 6 9 end{pmatrix} . What is the original vector mathbf{v} ?

[ ] A: begin{pmatrix} 3 3 end{pmatrix}
[ ] B: begin{pmatrix} 1 2 end{pmatrix}
[ ] C: begin{pmatrix} 6 3 end{pmatrix}
[ ] D: begin{pmatrix} 12 27 end{pmatrix}

Q6. Consider a linear transformation T represented by the matrix A = begin{pmatrix} 2 & 1 1 & 3 end{pmatrix} . Let P = begin{pmatrix} x y end{pmatrix} be a point in the plane. The transformation maps P to P' . If the area of the region transformed by T from a unit square in the xy -plane is A , and the determinant of A is detA , find the value of detA if the area of the transformed region is 5 .

[ ] A: 5
[ ] B: 2
[ ] C: 1
[ ] D: 10

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Q7. Consider the transformation grid below. If point A has coordinates 1, 2 , what are the coordinates of its image, A', after the transformation shown?

[ ] A: -4, 2

[ ] B: -2, 1

[ ] C: 2, -4

[ ] D: 1, -2

Q8. If A=begin{bmatrix}-4&-30&0end{bmatrix} and B=begin{bmatrix}-4&1-3&-2end{bmatrix} , find A+B .

[ ] A: begin{bmatrix}-8&-2-3&-2end{bmatrix}

[ ] B: begin{bmatrix}0&-43&2end{bmatrix}

[ ] C: begin{bmatrix}-8&-1-3&-2end{bmatrix}

[ ] D: begin{bmatrix}-8&-2-4&-2end{bmatrix}